

WASP-177b

R-2204

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-177_201111 · created 2026-08-14T00:59:26+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2459164.7062 +/- 0.0059 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.167 +/- 0.044
Transit depth (Rp/Rs)^2	2.8 +/- 1.48 [%]
Orbital Inclination (inc)	84.73 +/- 2.3
Airmass coefficient 1 (a1)	1.001 +/- 0.0096
Airmass coefficient 2 (a2)	0.0 +/- 0.014
Scatter in the residuals of the lightcurve fit is	0.96 %
Best Comparison Star	#3 - [524, 112]
Optimal Method	PSF photometry
Transit Duration (day)	0.048 +/- 0.028

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.167 ± 0.044 vs 0.1618 (deviation 0.12σ)
Duration fit vs published	0.048 d vs 0.0721 d (deviation 0.86σ)
Mid-time O–C	12.5 min
Depth SNR	3.8
Residual scatter	0.96 %

Light curve

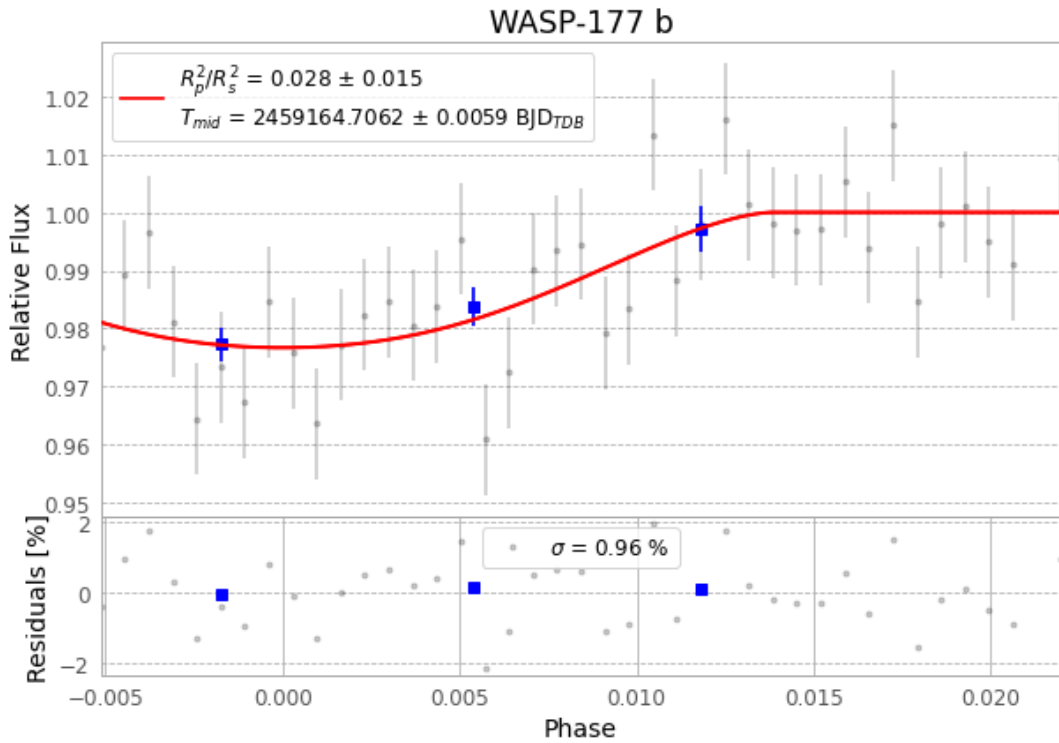


Figure 1 — FinalLightCurve_WASP-177 b_2020-11-10.png (SHA-256 0219a11e14744ef0...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [280, 239], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 c6239e33c06929cd...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

41 FITS frames (27 MB), WASP-177201111043014.FITS → WASP-177201111063012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-177-201111.log (e26355616da0...), FinalLightCurve_WASP-177 b_2020-11-10.png (0219a11e1474...), FinalLightCurve_WASP-177 b_2020-11-10.csv (479afa42130e...)

Compute resources

Wall time 87.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-14 00:59Z.